DISCRETE STRUCTURES

# CMSC250

## Chapter 17: Introduction to Sets

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Membership, subsets, diagrams, and set operations



# Contents

**1 Why sets matter** **3**

**2 Ways to specify a set** **3**

**3 Number sets and interval notation** **3**

**4 Subsets and equality** **3**

**5 Union, intersection, and difference** **4**

**6 Complements and the universe** **4**

**7 Power sets and cardinality** **4**

**8 Cartesian products** **5**

**9 Practice questions and complete solutions** **5**

**10 Common mistakes** **5**

# 1 Why sets matter

A **set** is a collection of distinct objects called **elements**. Sets describe possible inputs to functions, outcomes of experiments, and collections of mathematical objects. Order and repetition do not matter: $\{1, 2, 2, 3\} = \{3, 2, 1\}$. A set is determined by its members, not by how it is written.

**Key idea.** $x \in A$ means $x$ is an element of $A$; $x \notin A$ means it is not. $A \subseteq B$ means every element of $A$ is also in $B$. An element and a subset are different types of objects.

### Worked example: membership versus containment

**Question.** Let $A = \{1, \{2\}, 3\}$. Decide whether $2 \in A$, $\{2\} \in A$, and $\{2\} \subseteq A$.

**Solution.** $2 \notin A$, since 2 is not listed as a standalone member. $\{2\} \in A$ because the set $\{2\}$ itself is a member. But $\{2\} \nsubseteq A$ because its only member, 2, is not in $A$. These distinctions are common sources of mistakes.

# 2 Ways to specify a set

**Roster notation** lists elements: $A = \{2, 4, 6, 8\}$. **Set-builder notation** specifies a property: $A = \{2k : k \in \mathbb{Z},\ 1 \le k \le 4\}$. The **empty set** $\varnothing$ has no elements; $\{\varnothing\}$ is a nonempty set with one element.

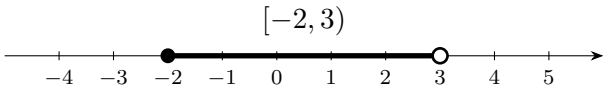
### Worked example: translate set-builder notation

**Question.** List $B = \{n \in \mathbb{Z} : -3 \le n < 4,\ n \text{ is odd}\}$.

**Solution.** The eligible integers are $-3, -2, -1, 0, 1, 2, 3$. Selecting odd values gives $B = \{-3, -1, 1, 3\}$. Always check the domain and strict versus non-strict inequalities.

# 3 Number sets and interval notation

We commonly write $\mathbb{N}$ for natural numbers (the convention about 0 varies), $\mathbb{Z}$ for integers, $\mathbb{Q}$ for rational numbers, and $\mathbb{R}$ for real numbers. For real intervals, $[a, b]$ includes endpoints while $(a, b)$ excludes them.




The filled endpoint includes $-2$; the open endpoint excludes 3. For integers, $\{n \in \mathbb{Z} : -2 \le n < 3\} = \{-2, -1, 0, 1, 2\}$.

# 4 Subsets and equality

By definition $A \subseteq B$ exactly when $\forall x\,(x \in A \Rightarrow x \in B)$. Equality is established by proving both $A \subseteq B$ and $B \subseteq A$. Every set is a subset of itself, and $\varnothing \subseteq A$ for every set $A$.

### Worked example: a two-direction subset proof

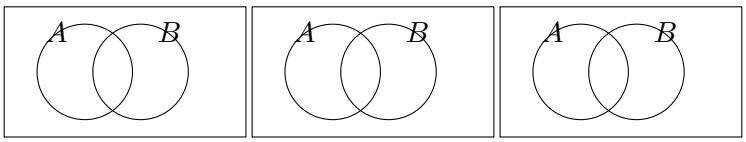
**Question.** Prove $\{2k : k \in \mathbb{Z}\} = \{n \in \mathbb{Z} : n \text{ is even}\}$.

**Proof.** Call the left set $A$ and right set $B$. First let $x \in A$ be arbitrary. Then $x = 2k$ for some $k \in \mathbb{Z}$, so $x$ is even and $x \in B$. Thus $A \subseteq B$. Conversely let $x \in B$ be arbitrary. By the definition of even, $x = 2k$ for some $k \in \mathbb{Z}$, so $x \in A$. Thus $B \subseteq A$. Hence $A = B$. ■

## 5 Union, intersection, and difference

For sets $A, B$, union means *or*, intersection means *and*, and difference means *in the first but not the second*:

$$A \cup B = \{x : x \in A \vee x \in B\}, \qquad A \cap B = \{x : x \in A \wedge x \in B\},$$
$$A \setminus B = \{x : x \in A \wedge x \notin B\}.$$




$A \cup B$ $A \cap B$ $A \setminus B$

*The outlines indicate the two overlapping sets; to evaluate a region, test whether an element satisfies the corresponding membership rule.*

### Worked example: compute operations explicitly

**Question.** Let $A = \{1, 2, 4, 6\}$ and $B = \{2, 3, 6, 7\}$. Find $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.

**Solution.** $A \cup B = \{1, 2, 3, 4, 6, 7\}$; $A \cap B = \{2, 6\}$; $A \setminus B = \{1, 4\}$; $B \setminus A = \{3, 7\}$. Notice difference is not commutative.

## 6 Complements and the universe

The complement $A^c$ is meaningful relative to a declared universal set $U$: $A^c = U \setminus A$. If $U = \{1, 2, 3, 4, 5\}$ and $A = \{1, 3, 5\}$, then $A^c = \{2, 4\}$. Without $U$, a complement can be ambiguous.

## 7 Power sets and cardinality

The **power set** $\mathcal{P}(A)$ is the set of all subsets of $A$. For $A = \{a, b\}$, $\mathcal{P}(A) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$. If $A$ has $n$ elements, each element has two choices (included/excluded), so $|\mathcal{P}(A)| = 2^n$.

### Worked example: power set of a three-element set

**Question.** Write all subsets of $C = \{1, 2, 3\}$.

**Solution.** Group by size: $\varnothing$; $\{1\},\{2\},\{3\}$; $\{1,2\},\{1,3\},\{2,3\}$; and $\{1,2,3\}$. There are $1+3+3+1 = 8 = 2^3$ subsets.

## 8 Cartesian products

$A \times B = \{(a,b) : a \in A,\ b \in B\}$ is the set of *ordered pairs*. For $A = \{1,2\}$, $B = \{x,y\}$, the product is $\{(1,x),(1,y),(2,x),(2,y)\}$. In general $|A \times B| = |A||B|$ for finite sets. Usually $A \times B \neq B \times A$.

## 9 Practice questions and complete solutions

**Question.** 1. Is $\varnothing \in \{\varnothing, 1\}$? Is $\varnothing \subseteq \{\varnothing, 1\}$?

**Solution.** Both are true. The first holds because $\varnothing$ is explicitly listed as an element; the second holds because the empty set is a subset of every set. **Question.** 2. Let $A = \{1,2\}$, $B = \{2,3\}$, and $U = \{1,2,3,4\}$. Find $(A \cup B)^c$.

**Solution.** $A \cup B = \{1,2,3\}$, so $(A \cup B)^c = \{4\}$ relative to $U$. **Question.** 3. Find $\mathcal{P}(\varnothing)$ and $\mathcal{P}(\{\varnothing\})$.

**Solution.** $\mathcal{P}(\varnothing) = \{\varnothing\}$ has one member. $\mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$ has two members.

## 10 Common mistakes

Do not confuse $\in$ with $\subseteq$; do not treat $\varnothing$ and $\{\varnothing\}$ as equal; always specify the universe for complements; and remember that Cartesian pairs are ordered.